

Olesia Dogonasheva

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GENERAL INFORMATION	I work on computational modeling of speech processing in the human brain. My work integrates Bayesian inference, predictive coding, and stochastic dynamical systems to formulate testable hypotheses, build mechanistic models, and interpret neural and behavioral data.	
EDUCATION	École normale supérieure, Doctorale Cerveau, cognition, comportement (ED3C), Paris PhD	2024
	HSE University, Institute for Cognitive Neuroscience, Moscow PhD	2023
	Alferov University, Nanobiotechnology department, Saint-Petersburg MD in Physics	2020
	ITMO University, Department of higher mathematics, Saint-Petersburg BD in Applied Mathematics and Computer Science	2015
COMPLEMENTARY EDUCATION	Bioinformatics Institute, Saint-Petersburg Statistics, data analysis, machine learning, bioinformatics tools	2020
	Computer Science Center, Saint-Petersburg Algorithms and data structures, C++	2012
PUBLICATIONS	D. Radushev, O. Dogonasheva, B. Gutkin, D. Zakharov “Dynamic states in a network of type-I Morris-Lecar neurons characterized using the Metric Framework”, <i>Chaos: An Interdisciplinary Journal of Nonlinear Science (Vol.36, Issue 5)</i>, 2026. D. Radushev, O. Dogonasheva, B. Gutkin, D. Zakharov “Metric framework of coherent activity patterns identification in spiking neuronal networks”, <i>Chaos, Solitons & Fractals: the interdisciplinary journal of Nonlinear Science, and Nonequilibrium and Complex Phenomena (in press)</i>, 2026. O. Dogonasheva, D. Zakharov, A.L. Giraud, B. Gutkin “Neuro-oscillatory models of cortical speech processing”, <i>Neural Networks</i>, 2026. O. Dogonasheva, D. Zakharov, K. Doelling, A.L. Giraud, B. Gutkin “Rhythm-based hierarchical predictive computations support acoustic–semantic transformation in speech processing”, <i>Nature Computational Science</i>, 2025. V. Tiselko, O. Dogonasheva, A. Myshkin, D. Zakharov, O. Valba “Interaction of functional brain networks is associated with k-clique percolation in the human structural connectome”, <i>Human Brain Mapping</i>, 2025. O. Dogonasheva, D. Radushev, B. Gutkin, D. Zakharov, “Dynamical manifold dimensionality as characterization measure of chimera states in bursting neuronal networks”, <i>Communications in Nonlinear Science and Numerical Simulation</i>, 108321, ISSN 1007-5704, 2025 O. Dogonasheva, B. Gutkin, D. Zakharov “Cluster formation in modular pyramidal-interneuron gamma networks under spike-frequency adaptation”, <i>The European Physical Journal Special Topics</i>, p. 1-15, 2025 O. Dogonasheva, E. Postnikov, A. Lavrova “Shaping spiking patterns through synaptic parameters as revealed by conventional and wavelet-based bifurcation analysis”, <i>The European Physical Journal Special Topics</i>, p. 1-13, 2023 O. Dogonasheva, D. Kasatkin, B. Gutkin, D. Zakharov “Multistability and evolution of chimera states in a network of type II Morris-Lecar neurons with asymmetrical nonlocal inhibitory connections”, <i>Chaos: An Interdisciplinary Journal of Nonlinear Science</i>, vol. 32, no. 10, p. 101101, 2022 O. Dogonasheva, D. Kasatkin, B. Gutkin, D. Zakharov “Robust universal approach to identify travelling chimeras and synchronized clusters in spiking networks”, <i>Chaos, Solitons and Fractals: the interdisciplinary journal of Nonlinear Science, and Nonequilibrium and Complex Phenomena</i>, vol. 153, p. 111541, 2021	

PREPRINTS	O. Platonova, O. Dogonasheva, A.L. Giraud, S. Bouton “Contextual Prediction Tunes the Tempo of Speech Segmentation”, <i>bioRxiv</i> , 2026-03, 2026.	
INVITED TALKS	Transmission de l’information visuelle, Institut de la Vision, “Rhythm-based hierarchical predictive computations support acoustic–semantic transformation in speech processing”	2026
	University Claude Bernard Lyon 1, “Rhythm-based hierarchical predictive computations support acoustic–semantic transformation in speech processing”	2025
	Computational Neuroscience Group, Université de Bordeaux, “Computational modelling of speech processing in the Auditory Cortex”	2025
	The Laboratory of Theoretical Physics and Statistical Models (LPTMS), University Paris-Saclay, “Delta-theta-gamma interplay for contextualized speech parsing”	2023
	LaCNS Group, the Donders Centre for Cognitive Neuroimaging, Radboud University, “Brain-Rhythms-Based Inference (BRyBI) model for Contextualized Speech Parsing”	2023
	Institute of Radio-engineering and Electronics RAS, Saratov, Russia, “Travelling chimera states in the ring of II-type Morris-Lecar neurons with asymmetric nonlocal inhibitory connections”	2020
MAIN CONFERENCES	O. Dogonasheva, B. Gutkin, A.L. Giraud “Rhythm-based neuronal inference supports robust and efficient speech processing”, <i>Cosyne</i> , Lisbon, Portugal, Mar 2026. O. Dogonasheva, A.L. Giraud, B. Gutkin “Rhythm-based neuronal inference supports robust and efficient speech processing”, <i>ICMNS</i> , Barcelona, Spain, Jun 2025. O. Dogonasheva, D. Zakharov, A.L. Giraud, B. Gutkin “Brain-Rhythm-based Inference (BRyBI) for time-scale invariant speech processing”, <i>Bernstein</i> , Frankfurt, Germany, Sep 2024. O. Dogonasheva, O. Platonova, D. Zakharov, A.L. Giraud, B. Gutkin “Rhythmically structured predictive coding enables invariant semantic recovery”, <i>Cosyne</i> , Lisbon, Portugal, Feb 2023. O. Dogonasheva, D. Zakharov, A.L. Giraud, B. Gutkin “Hierarchical Oscillation as a Mechanisms for Speech Parsing”, <i>Bernstein</i> , Berlin, Germany, Sep 2023. O. Dogonasheva, D. Zakharov, A.L. Giraud, B. Gutkin “Delta-theta-gamma interplay for contextualized speech parsing”, <i>32nd Annual Computational Neuroscience Meeting</i> , Leipzig, Germany, Jul 2023. O. Dogonasheva, B. Gutkin, D. Zakharov “Correlation dimension as a measure of chimera states”, <i>NoDycon</i> , Rome, Italy, Jun 2023. O. Dogonasheva, D. Zakharov, B. Gutkin “Brain-Rhythm-based Inference (BRyBI) for time-scale invariant speech processing”, <i>Cosyne</i> , Montreal, Canada, Mar 2023. O. Dogonasheva, B. Gutkin, D. Zakharov “Identification of multistable coherent regimes in spiking neural networks”, <i>Dynamics Day Europe</i> , Aberdeen, Scotland, UK, Aug 2022. O. Dogonasheva, B. Gutkin, D. Zakharov “Multistability of coherent states in ring networks of type II neurons with asymmetrical nonlocal inhibitory connectivity”, <i>30th Annual Computational Neuroscience Meeting</i> , Jul 2021.	
PATENTS	Patent for a computer program No. 2022684705, Software package for the identification of dynamical regimes in spiking neural networks	2022
	Patent for a computer program No. 2022684101, Software package for calculation of the chimera state speed in spiking neural networks	2022
	Patent for a computer program No. 2021616670, Software package for automatic simulation of chemical reactions networks during the deposition of molecular coatings on surfaces	2021
	Patent for a computer program No. 2020615220, Software package ReaNet for automatic modeling of thermal chemical and photochemical reactions	2020

AWARDS & SCHOLARSHIPS	Vernadsky Fellowship Award, Ambassadeur de France en Russie	2021-2024
	Idea Foundation Fellowship, Research Center "IDEAS"	2021-2023
	Stipendium, Freien Universität Berlin, Germany	2020
	Winner of the competition "Start - Digital Technology", Innovation Promotion Fund, Moscow, Russia	2020
	Prizewinner in the "International gemini Mars design competition", The Mars society, USA	2016
SCHOOLS & INTERNSHIPS	NeuroDecoder, Workshop on Advanced Methods for Neural Decoding of Speech and Auditory Signals	2025
	Junior Scientists Workshop on Recent Advances in Theoretical Neuroscience (smr 3943), Workshop on Computational neuroscience in Trieste, Italy	2024
	Bio-inspired deep learning, Workshop on Simulation-Based Inference (SBI) in Mainz, Germany	2024
	Neurobridges, Scientific School in Cluny, France	2022
	Mathematics, Theoretical Physics and Mathematical Methods of Data Analysis in Neuroscience, Scientific School in Sirius Mathematics Center, Sochi, Russia	2021
	Neuromatch Academy, Summer school in Computational Neuroscience	2020
	Bioinformatics Summer School, MIPT, Moscow, Russia	2019
TEACHING EXPERIENCE	HSE University, Moscow - Advanced Calculus, assistance - Theoretical neuroscience, seminars - Brain and Mind, seminars	2021 – 2022
	ITMO University, Saint-Petersburg - Higher mathematics, 3-terms course - Probability theory and mathematical statistics, 2-terms course	2014 – 2017
RESEARCH EXPERIENCE	Postdoctoral Researcher, Institut de l'Audition, Pasteur Institute , Paris Study brain rhythms influence on speech coding	2024 – now
	Researcher, Group for Neural Theory, LNC2, École Normale Supérieure , Paris Study brain rhythms influence on speech coding	2022 – 2024
	Junior researcher, CCDM, Higher School of Economics , Moscow Study synchronization processes and chimera states in spiking neural networks	2020 – 2023
	Project at Skoltech , Moscow Discovery of allosteric binding site in G protein-coupled receptors	2020
	Project at Bioinformatics Institute , Saint-Petersburg Phenome-wide functional analysis of human genetic association data	2019
	Assistant, Nanobiotechnology laboratory, Alferov University , Saint-Petersburg - Development of optimization algorithms to automatic search of chemical reaction pathways - Implementation of a genetic algorithm for the amino acid substitutions selection in the chromophore to increase the absorption coefficient	2018 – 2020

- Implementation of machine learning algorithms to optimize force field parameters

Assistant, Department of higher mathematics **ITMO University**, Saint-Petersburg

2014 – 2017

- Solving spectral problems for spiral-type quantum graphs

Technician, Mathematical modelling department, **Scientific Center for Robotics and Technical Cybernetics**, Saint-Petersburg

2014 – 2015

- Solution for the inverse kinematics problem for a six-chain robot manipulator
- Ice dynamics modelling as a part of the project of rescue robots on the oil platforms